5 Gas–Liquid Contacting Equipment

In Chapters 2–5, we discussed the basic principles of diffusion and mass transfer in a single phase or at a phase boundary. We also introduced the idea and strategy of contacting two phases in a stage-wise manner in order to transfer a solute from one phase to the other effectively and efficiently. Equipment and devices for interphase mass transfer can be broadly classified into the following categories:

- Gas–liquid or vapour–liquid contacting (gas absorption and stripping, distillation, humidification or dehumidification and water cooling)
- Liquid–liquid contacting (solvent extraction)
- Gas–solid contacting (drying and adsorption)
- Liquid–solid contacting (leaching, crystallization and ion exchange).

In many applications, the mass transfer equipment operates in the continuous mode. Batch contacting is sometimes used, particularly for low processing capacities or in small production units. The objective is to achieve an intimate contact between the phases and to ensure a high rate of mass transfer. A high degree of turbulent mixing is created to properly disperse one phase into the other. This helps in generating a large interfacial area of contact as well as in increasing the mass transfer coefficient. In a 'tray tower' (described below), for example, interfacial area is generated by passage of the gas (or vapour) into the liquid through perforations or openings; in a 'packed tower', on the other hand, interfacial area is generated by spreading the liquid on the packing surface. A gas–liquid contacting equipment may be provided with heating or cooling arrangements wherever necessary in order to maintain the desired process conditions. Mass transfer equipment is mostly 'custom-built' (i.e. designed and fabricated according to the requirements of the client; these types of equipment cannot be readily purchased from the market as 'on-the-shelf' items like pumps, valves, blowers, air-conditioners, etc.). The common types of equipment used for carrying out gas–liquid mass transfer in practice and their constructional and operational features are described in this chapter. Equipment for contacting other combinations of phases (liquid–liquid, gas–solid, etc.) mentioned above will be described in respective chapters.

5.1 INTRODUCTION TO GAS–LIQUID CONTACTING

A variety of gas–liquid contacting equipment is in use. In some of these, the gas is dispersed in the liquid in the form of bubbles (for example, tray towers, bubble columns, agitated vessels, etc.); in some others the liquid is dispersed in the form of droplets or discontinuous films in a continuous gas phase (for example, spray towers, packed towers, venturi scrubbers, etc.). Sometimes both gas and liquid phases are continuous (for example, a 'falling film contactor' which is used for certain gas–liquid reactions). Tray and packed columns are most widely used for gas–liquid contacting — namely for gas absorption, stripping, distillation. These are also used for liquid–liquid extraction (discussed in Chapter 8). The usual ranges of the two important mass transfer parameters, namely the mass transfer coefficient k_L and the specific interfacial area $\overline{a}$, attainable in common commercial gas–liquid contactors are given in Table 5.1.

Table 5.1 Common range of values of mass transfer parameters in commercial gas–liquid contactors

Contacting equipment	*Superficial gas velocity* (m/s at normal condition)	*Mass transfer parameters*		
		$k_L \times 10^4$ (m/s)	$k_y \times 10^3$ (kmol/s.m^2.Δy)	$\overline{a}$ (m^2/m^3)
Sieve tray	0.02– 0.5	1– 20	0.5–6.0	100–200
Packed column	0.1–1.2	1– 20	0.03–2.0	10–350
Bubble column	0.1–0.3	0.4–3.0	0.5–2.0	50–600
Spray column	0.05–3.0	0.7–1.5	0.5–2.0	10–100
Mechanically agitated contactor	0.05 –3.0	0.3 – 4.0		100–1000

5.2 TRAY OR PLATE COLUMN

A tray column primarily consists of a vertical cylindrical shell and a set of 'tower internals' that include (i) trays or plates on which the gas–liquid contact occurs, (ii) arrangements for flow of the liquid from one tray to the lower one through the downcomer, and (iii) inlet and outlet nozzles for the two phases. Figure 5.1 schematically shows a few essential parts of a 'sieve tray' column.

In a gas absorption application, the liquid enters the top tray through a nozzle. It impinges on a baffle plate, moves across the tray and flows into the lower tray through a 'downcomer'. The sieve trays shown in the figure are made of perforated metal sheets. (There are various other kinds of trays which will be described later.) The gas flows upwards and vigorously bubbles through the liquid on a tray, forming a turbulent 'gas–liquid dispersion' in which bubble breakage and coalescence occur continuously. An average depth of the dispersion is maintained on a tray. A large gas–liquid contact area and a high mass transfer coefficient are achieved. Mass transfer from the gas to the liquid (or from the liquid to the gas) phase occurs depending on the direction of the driving force. For example, in 'gas absorption', the solute gets transported from the gas to the liquid phase; the reverse occurs in stripping (discussed in Chapter 6). The gas then leaves the froth or dispersion (this is called 'gas–liquid disengagement') and enters the next upper tray. The liquid flows across a tray and then over a 'weir' to enter into the downcomer (the functions of a downcomer and a weir are discussed in Section 5.2.3). The downcomer is a region near the wall, separated by a 'downcomer plate', in which the bubbles get disengaged from the liquid.

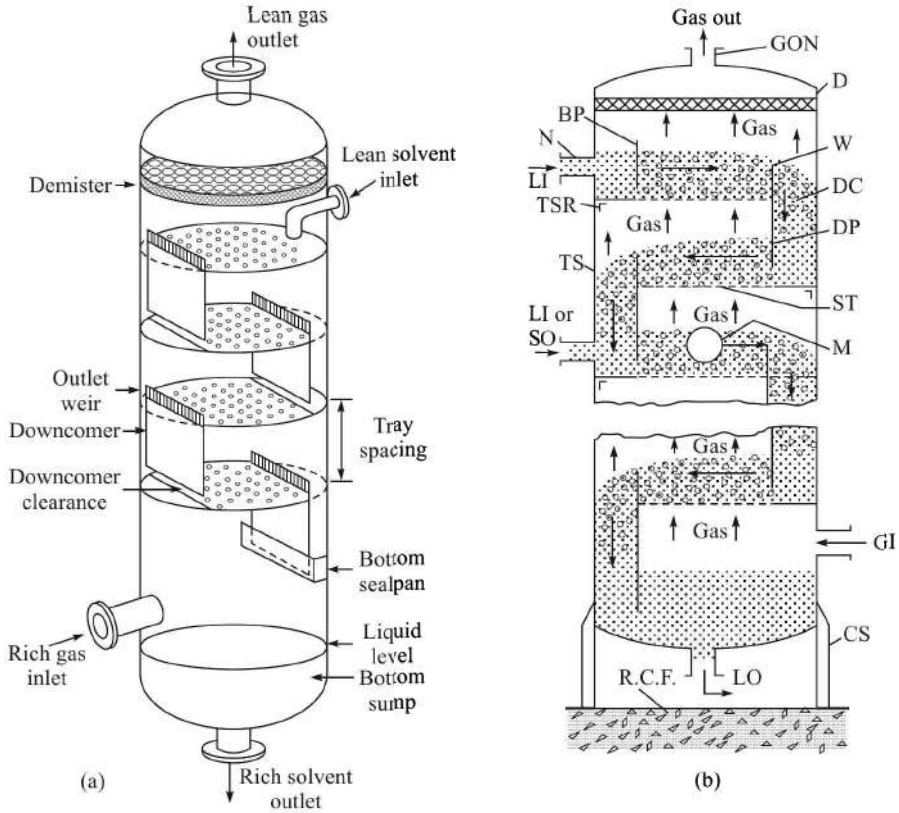


Figure 5.1 (a) Schematic of a 'sieve tray' column and (b) sectional sketch of a 'sieve tray' column [BP: baffle plate; CS: column support; D: demister; DC: downcomer; DP: downcomer plate; GI: gas inlet; GON: gas outlet nozzle; LI: liquid inlet; LI or SO: liquid inlet or side-stream outlet; M: manhole; N: nozzle: RCF: reinforced concrete foundation; ST: sieve tray: TS: tower shell; TSR: tray-support ring; W: weir].

The clear liquid flows to the next lower tray. In this way, countercurrent and stage-wise gas–liquid contact takes place in the tower. Each tray acts as a stage in which the liquid flowing down from the upper tray and the gas flowing up from the lower tray come into contact; the tower acts as a cascade (a 'stage' and a 'cascade' have been defined in Chapter 4).

The constructional and operational features of a tray tower and tower internals are briefly discussed below. Further details are available in the literature (see, for example, Kister, 1989; Ludwig, 1997; Chopey, 1997).

5.2.1 The Shell

The shell is usually made of a metal or an alloy. Plastic (for example, FRP) shells are also used sometimes. The material is selected on the basis of corrosiveness of the fluids, temperature and pressure conditions, and cost. If the shell diameter is small, it may be made of several flanged sections in order that the trays may be fitted into it and maintained, when necessary, by opening

it. Tray towers of diameter less than 1 metre are rarely used. On the other extreme, towers as big as 10 metre in diametre are known to be in use.

A metal pipe may be used as the shell if the required diameter of the tower is small. Larger diameter shells are made from metal sheets by rolling. In fact, smaller sections are first made and then welded to get the shell of required diameter and length. Shell thickness is calculated by using a standard vessel design code (for example, ASME Section VIII; IS 2825; Moss, 2004). A tower may have flanged or welded ends depending upon the diameter.

Trays are bolted to tray support rings. Metal angles (L-section) rounded and welded to the shell may act as the tray support rings (the support rings also act as 'stiffeners' of the shell). Big diameter trays need support beams in addition, in order to prevent sagging of a tray under liquid load. Channels and I-section beams clamped or bolted at both ends to the support rings are used to support trays as shown later in Figure 5.11.

A tower is generally 'skirt-supported' on concrete foundations (Moss, 2004). The support should be strong enough to take the column weight, the liquid load and the wind stress. The seismic factor at the particular location is also needed to be considered during column and support design. Since the bending moment due to wind load is maximum at the tower bottom, it is a common practice to use different wall thicknesses for different sections of the shell. The bottom section has the maximum thickness and the upper sections have gradually smaller thicknesses. This strategy substantially reduces the weight, and hence the cost, of the shell. A tall tower has a ladder fixed on its wall. Small platforms with railings are fixed to the shell wall at the level of each manway for easy and safe access into the tower. Towers are insulated against heat exchange with the surroundings. A tower is erected on the foundation using a crane. Examples of erection after lifting a tower by a helicopter have been reported.

5.2.2 The Tray

A 'tray' has two major functions:

1. It allows the gas to flow through the holes or passages; the gas vigorously bubbles through the liquid to form a 'gas–liquid dispersion'. The tray holds the dispersion on it.
2. The trays separate the column into a number of compartments each of which constitutes a stage. Mass transfer between the phases occurs on a tray. Therefore, the trays as a whole constitute the heart of a column. The performance of a column depends upon the performance of the trays.

Trays are also called 'plates'. There are quite a few types of trays in use. The *bubble-cap tray* will be described first.

The bubble-cap tray

This is the oldest type of tray and is still in commercial use. A bubble cap consists of two major components—a bell-shaped 'cap' and a 'riser' (also called a 'chimney'). Figure 5.2(a) and (b) shows a typical bubble-cap design. The riser is inserted through a hole on the tray floor (or deck) and the bell-shaped cap is bolted to it. A ring gasket (not shown) is used below the nut. The riser or chimney is a piece of tube with a flared or expanded bottom end (there are other designs too). In fact, the riser acts as the vapour passage and also holds the cap.

A bubble cap has slots on its wall. The shape of the slots may be rectangular (most common), triangular, trapezoidal, or saw-tooth type. In the bell-shaped cap shown in Figure 5.2(a) and 5.2(b), the slots end a little above the bottom of the cap. The continuous portion of a cap below the slots is called the 'shroud ring'. The cap may be mechanically weak if there is no shroud ring and the slots extend up to the bottom. A bell-shaped cap rests on three short legs or tabs, integral to the cap, placed 120° apart. The open region thus provided between the tray floor and the cap bottom is called the 'skirt clearance'. Such a design helps in reducing accumulation of sediments and provides an additional area for gas flow if there is a sudden surge of the gas. Figure 5.2(c) shows a few other designs of the bubble cap.

The caps and the risers are made of low carbon steel, stainless steel or any other suitable material that can withstand the environment within the tower. Caps are arranged on a tray on equilateral triangular pitch with rows normal to the direction of liquid flow. Bubble caps generally range from 1 inch to 6 inches in diameter. The rectangular slots may be of the following size range:

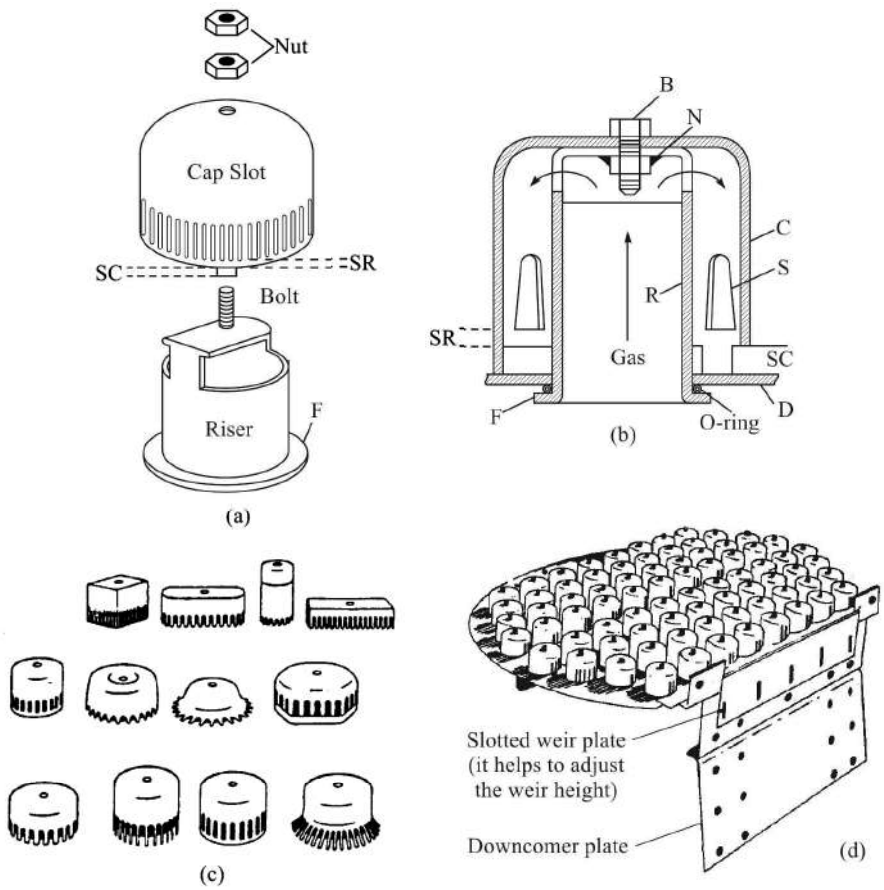


Figure 5.2 Bubble caps and a bubble-cap tray: (a) an exploded view of a bubble cap and (b) sectional diagram of a bubble cap [B: bolt; C: cap; D: tray deck; F: flared bottom of the riser; N: nut; R: riser; S: slot; SC: skirt clearance; SR: shroud ring] *Courtesy:* ACS Industries, USA; (c) different types of bubble-caps and (d) a bubble-cap tray

length ½ inch to 1½ inches, width ⅛ to ⅜ inch. Quite a few other slot geometries are used as well [Figure 5.2(c)]. The number of slots in a cap ranges from 12 to 70. For a medium-size column (4 to 10 feet), 4-inch caps are common. The cap-pitch (centre-to-centre distance) is about 1 inch to 2 inches more than the cap outer diameter. A bubble-cap tray is shown in Figure 5.2(d).

The sieve tray

This is the simplest type of tray in which the bubble caps are replaced by holes or perforations for entrance of the gas into the liquid. The holes are of relatively small diameter — usually ranging from ⅛ to ½ inch. This is why the name 'sieve tray' (also called 'perforated tray'). For clean services, use of a hole diameter of 3/16 inch is common. However, for liquids that foul or cause deposition, a hole diameter of ½ inch may have to be used. In vacuum services, ⅛ inch hole diameter is preferred. Small holes enhance tray capacity, reduce entrainment, reduce weeping, promote froth regime operation and exhibit better mass transfer characteristics.

Holes are made by punching rather than by drilling a tray sheet (small holes may have to be made by drilling) in order to reduce the labour cost. These are arranged on an equilateral triangular pitch or a square pitch. The pitch (i.e. the centre-to-centre distance between the two adjacent holes) varies between $2.5d_H$ and $5d_H$ ($3.8d_H$ is common), where d_H is the hole diameter [as shown marked on Figure 5.12(b)]. The rows of holes are oriented normal to the direction of liquid flow. A sieve tray typically has a free area of 6 to 10% [free area fraction = hole area/tower cross-section, less the downcomer area]. The layout of a sieve tray, the downcomers and the weirs are shown in Figure 5.3(a) and (b). The assembly of a sieve tray is shown in Figure 5.3(c).

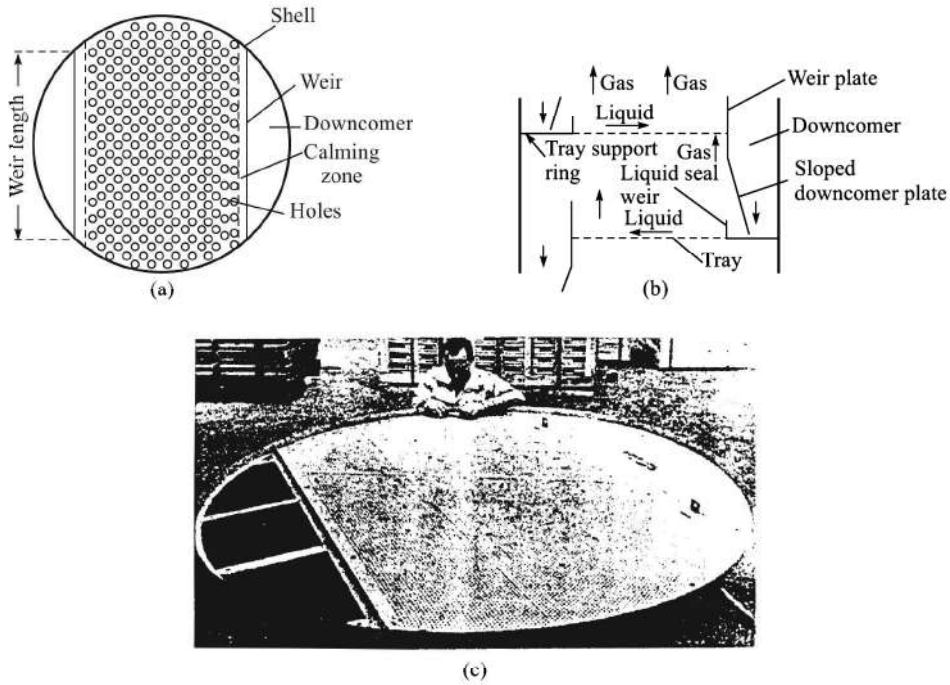
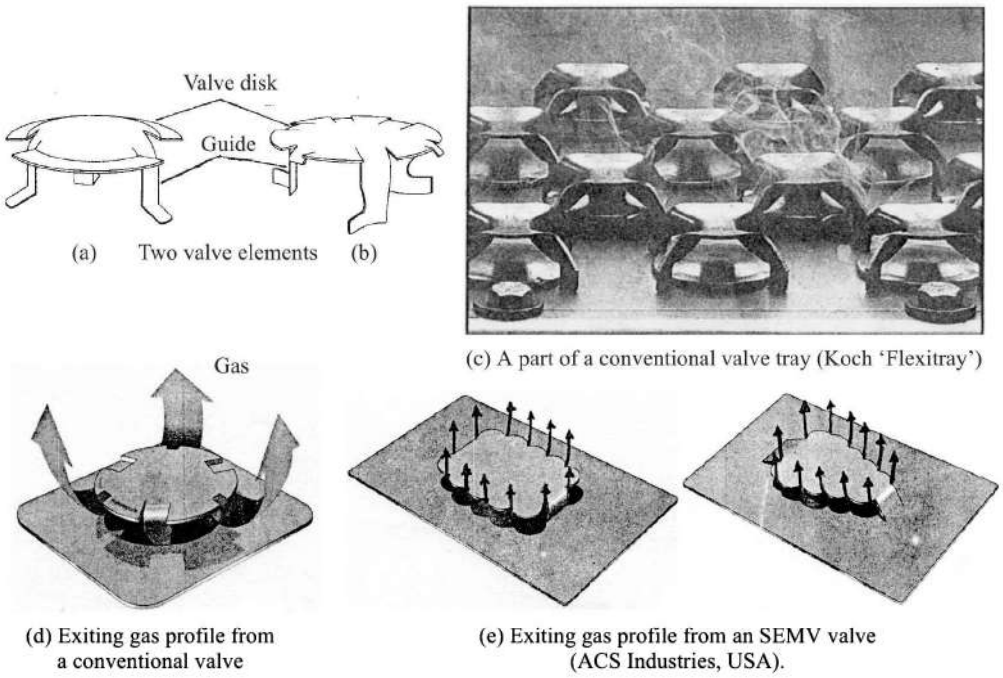


Figure 5.3 (a) Sieve tray layout, (b) downcomer, weir and the liquid seal weir, and (c) assembly of a sieve-tray.

The valve tray

The valve tray is a relatively new class of tray that provides variable area for the gas or vapour flow depending upon the flow rate or 'throughput'. This is why it is called 'valve tray'. A valve tray is a proprietary (that is patented) tray. Different types of valve trays are made by different companies.

A common valve tray has sufficiently large punched holes on the tray floor, each fitted with a movable disk, generally circular. A disk has guides that can slide vertically up or down along the thickness of the tray floor [Figure 5.4(a) and (b)]. The opening for the gas flow changes in this way, but the disk is always held in the same vertical line. As the gas flow increases, the disk is automatically raised. It settles down at a low vapour rate to prevent 'weeping'. The guides or retaining legs are bent at the end so that the disk does not pop up or gets detached even at a large vapour rate. The valve, a part of a valve tray (Koch Flexitray having 'caged valves') and the vapour flow profile through valves are all also shown in Figure 5.4(c). A valve tray offers a high 'turn down' ratio (this term will be defined later).



(c) A part of a conventional valve tray (Koch 'Flexitray')

(d) Exiting gas profile from a conventional valve

(e) Exiting gas profile from an SEMV valve (ACS Industries, USA).

Figure 5.4 A few types of valves, a part of a valve tray, and vapour flow profile through valves.

Since these are proprietary trays, the details of their design are not available in the open literature. These trays are extremely flexible because of the variable area of gas flow. The trays do not easily acquire deposits from dirty liquids, polymers or other solids because of the up and down motion of the disk and the guides. Valve units are, therefore, self-cleaning. The valve tray is a good choice for highly fouling services. In addition, they offer lower pressure drop than the bubble-cap type and generally they are cheaper than the latter type. A few problems common to

all kinds of valve trays are (i) mechanical wear and corrosion because of continuous movement of the valve legs, and (ii) sticking of the disk on the tray if there is sticky deposition on the tray.

Excessive opening of the valves at a low gas rate causes weeping and valves should therefore be heavy enough to prevent this. On the other hand, heavy valves incur a greater pressure drop. As a compromise, alternate rows of heavy (made from 12 or 14 gauge sheets) and light (made from 16 gauge sheets) valves are used. In a common valve tray layout, 12 to 16 valves per ft^2 of tray area are accommodated. Common orifice and disk diameters are of the order of 1½ and 2 inches. The disks typically rise $3/16$ to $7/16$ inch above the tray deck. The flow area of fully open valves on a tray is about 10–15% of the active tray area (the active tray area is the tray cross-section less the downcomer area).

Because of high flexibility, high turndown ratio and relatively low cost, valve trays are now widely used for gas absorption and distillation. Various valve designs from different companies are available. Koch-Glitch makes twenty types of valves to suit a variety of services. In new installations, bubble-cap trays are rarely used; sieve trays and the valve trays are the preferred tray types. In contrast to movable or 'floating' valves described before, many companies have come up with 'fixed' valves that have properly shaped and profiled fixed covers above valve opening for better gas/vapour distribution. Typical fixed valves are shown in Figure 5.4(e) here and later on in Figure 5.14(c).

Other types of trays

There are a few other less common trays used for gas–liquid contacting. Typical examples are: Dual flow tray, Ripple tray, Leva film tray, etc. (Billet, 1979). A dual flow tray is a sieve tray without downcomers. The vapour (or gas) bubbles through the liquid and the liquid simultaneously weeps through the perforations (this is why it is called 'dual flow'). An average froth height, which is less than that on a standard sieve tray, is maintained. Since the contact time between the liquid and the vapour is small, the efficiency of a dual flow tray is less than that of a sieve tray. However, this is partly compensated by the increased active area (there is no loss of area due to the downcomers). Dual flow trays are known to be in use in a limited number of installations.

The Ripple tray (supplied by Stone and Webster) is a variation of the dual flow tray (Sloley, 1999). The tray has a corrugated deck and the liquid preferentially weeps through the troughs on the deck.

5.2.3 Downcomers and Weirs

The 'downcomer' is a passage through which the liquid flows down from one tray to the next below (Figures 5.1 and 5.3). The desired depth of the gas–liquid dispersion is maintained on a tray by using a 'weir' in the form of a vertical plate. The liquid, along with some dispersed gas or vapour bubbles, overflows the weir and enters the 'downcomer' or the 'downspout'. Disengagement of the gas as bubbles occurs in the upper region of a downcomer. The lower region contains clear liquid that enters the lower tray. The downcomer must provide sufficient residence time for gas–liquid disengagement (the residence time is the ratio of the downcomer volume and the volumetric flow rate of the clear liquid). The residence time is usually 3 to 5 seconds. However for a foaming liquid, considerably higher residence time and therefore a larger downcomer volume has to be provided. The 'clear liquid' velocity in the downcomer normally ranges between 0.3 and 0.5 ft/s. This value may vary depending upon the liquid properties.

A 'downcomer plate†' may be straight or inclined. The downcomer shown in Figure 5.1 or in Figure 5.7 is a straight plate which covers a constant segment of the tower cross-section. It is called a 'segmental downcomer'. In the downcomer design shown in Figure 5.3(b), the lower end of the downcomer plate is bent towards the column wall. This design, called 'inclined' or slopped downcomer, provides more downcomer space in the upper region where the gas–liquid disengagement occurs but less space in the lower region where the liquid is rather clear. In another variation, called 'stepped downcomer', the downcomer area decreases in steps [Figure 5.5(a)]. The downcomer plate is fixed to the tray from which the liquid overflows. The weir plate is bolted to the downcomer plate through a slotted bolt hole, so that the weir height can be easily adjusted [see Figure 5.2(d)]. The shape of an inclined downcomer and the layout of the perforations on a sieve tray are schematically shown in Figure 5.3. The lower edge of the downcomer plate should remain dipped into the liquid on the lower tray maintaining a clearance, (called 'downcomer clearance') of ½ to 1 inch from the tray floor. This makes a 'downcomer seal' [also shown in Figure 5.5(b) as D.S.] prevent 'short-circuiting' of some of the gas through the downcomer. An alternative is a depressed downcomer seal [Figure 5.5(c)]. The depressed region of the tray is called the 'seal pan' or 'seal pot'.

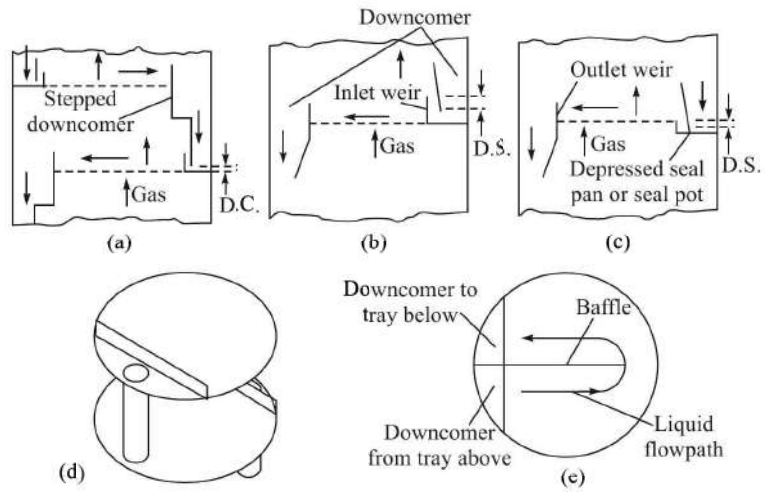


Figure 5.5 Downcomer types and seals: (a) a stepped downcomer, (b) a normal downcomer seal (D.S.), (c) a 'depressed' downcomer seal, (d) a 'pipe downcomer', and (e) the downcomer on a 'reverse-flow tray'. [D.C. = 'downcomer clearance'].

The weir length may vary from 60 to 80% of the tower diameter; the downcomer area (on one side) correspondingly varies from 5 to 15% of the tray area. A weir height of 1 to 2 inches is generally maintained. The row of bubble caps, valves or perforations (in case of a sieve tray) adjacent to the overflow weir should be about 3 to 4 inches away from it so that partial disengagement of the gas occurs before the liquid overflows the weir and enters the downcomer. This blank region on the tray deck is called the 'calming zone' [see Figure 5.3(a)]. The area of a tray that contains bubble caps, holes or valves is called the 'active tray area'.

† A 'downcomer plate' is just a metal plate and should not be confused with a tray.

In small columns, pipe downcomers ('downtubes') are preferred [see Figure 5.5(d)]. In a 'reverse-flow tray', the liquid entering a tray flows over half of the active tray area and then changes direction around a central baffle [see Figure 5.5(e)]. Both the downcomer and the downcomer seal are placed on the same side of such a tray. A longer liquid flow path is created this way. Such an arrangement is suitable if the liquid flow rate in the column is small.

5.2.4 Nozzles

A tower for contacting a liquid and a vapour (or a gas) should be provided with a few nozzles for feed entry (both gas and liquid), entry of reflux at the top and of the reboiler vapour return at the bottom (in a distillation column), and for product withdrawal from the tower. The primary criterion of a feed nozzle design is to ensure that the feed is introduced with minimum splashing or jetting (the velocity of liquid feed in the nozzle should not exceed 1 m/s). The feed should be evenly distributed and mixed with internal liquid or vapour. A few nozzle types for feed entry and liquid draw-off are shown in Figure 5.6. In case of a multipass tray, the feed or the reflux stream has to be proportionally balanced to the active areas of a tray.

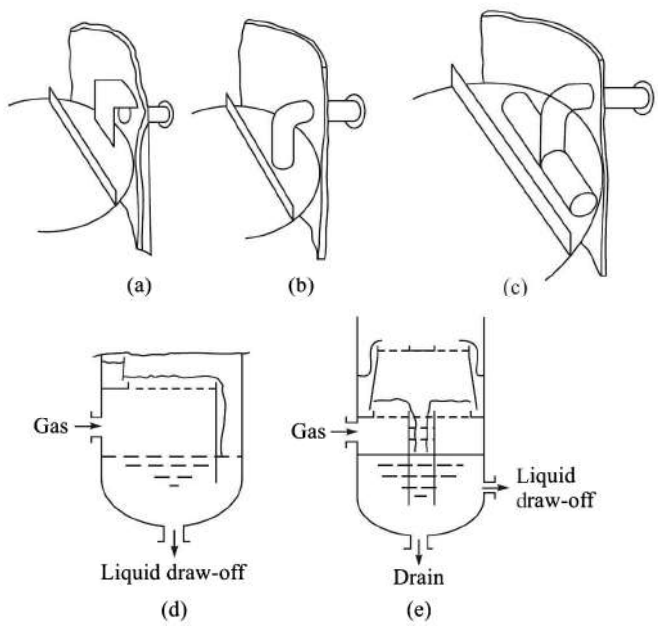


Figure 5.6 Feed entry and product draw-off nozzles: (a) an internal baffle prevents 'jetting' of the liquid feed, (b) a downward-bent feed nozzle, (c) a bifurcated feed nozzle, and (d) and (e) gas inlet and liquid draw-off nozzles [5.6(e) shows a split flow tray].

The location of the feed nozzle as per design calculations may not always be the optimum in practice. As a precautionary measure, one or two additional nozzles are sometimes provided above and below the calculated feed tray to provide some flexibility. The feed pipe may be connected to any of these adjacent nozzles if required for improved performance; otherwise these

are just kept closed. A few ports or small nozzles for sampling and for instrument (temperature and pressure sensors) connection are also provided at suitable locations for the measurement of composition, temperature, and pressure. The bottom feed or the reboiler vapour return nozzle must be placed above the bottom liquid level. Viewing ports are sometimes provided for frequent inspection and testing.

'Manholes' are provided for each of 10 to 15 trays for access of a workman into the column for inspection, cleaning, and maintenance. The manhole diameter is kept in the range 16 to 24 inches. Larger manholes may be necessary for admitting a person wearing protective clothing and a breathing apparatus depending upon the environment within a tower. For towers of less than 3 ft diameter, handholes (8 to 14 inches) are provided for the purpose of cleaning. A ladder and an access platform are needed to be built for all manholes 12 ft or higher from the floor. A manhole or a handhole is covered with a gasketed manhole cover which is nothing but a 'blind flange'.

5.2.5 Mist Eliminator

Even under normal operating conditions, a little entrainment of liquid in the upflowing vapour may occur. In order to prevent entrainment in the vapour leaving the top tray, a pad made of wire mesh or a pack of suitably bent and spaced thin sheets is fitted above it. The droplets are retained after they strike the surface of the pad (retention by 'inertial impaction'). Such a device is called 'mist eliminator' or a 'demister' (shown in Figure 5.31).

5.3 OPERATIONAL FEATURES OF A TRAY COLUMN

The flow phenomena on a tray with the gas bubbling vigorously through the flowing liquid are pretty complex. The tray internals are selected and designed, keeping in view the complexity of the flow and the problems that may arise out of it.

5.3.1 Hydraulic Gradient and Multipass Trays

The difference between the 'clear liquid heights' at the points of inlet and outlet on a tray is called the 'hydraulic gradient' or 'liquid gradient'[††] [Figure 5.7(a)]. It is the liquid head required for overcoming the resistance to liquid flow on a tray. The hydraulic gradient on a tray should not exceed a fraction of an inch. It should preferably be kept within ½ inch. An excessive liquid gradient causes severe malfunctioning of the tray; most of the gas flows through the holes near the middle of the tray and at the outlet weir (where the 'effective liquid depth' on the tray is less) and only a small part flows through the holes at the liquid inlet side of the tray. Such maldistribution of the gas or the vapour (also called 'vapour channelling') severely reduces the 'tray efficiency'. In extreme cases, liquid 'dumping' or 'back-trapping' may occur through the end where the liquid enters the tray [Figure 5.7(b)]. Hydraulic gradient is a very important quantity to be checked during tray design. It remains pretty small for a sieve tray. But for a bubble-cap tray it may be significant because the bubble caps offer a larger resistance to liquid flow.

[††] The term gradient appears to be a misnomer. 'Gradient' means the rate of change of a quantity with position. But the hydraulic gradient is really the 'difference' of liquid heights.

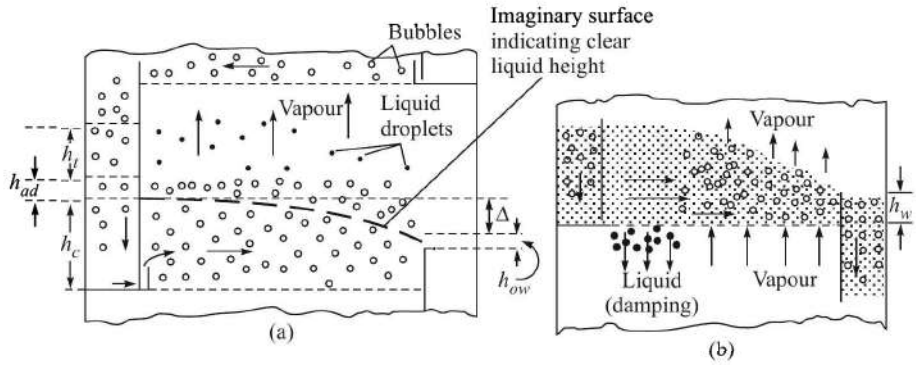


Figure 5.7 (a) Hydraulic gradient and clear liquid backup in the downcomer: h_c = clear liquid height at the tray inlet, h_{ad} = head loss for flow below the 'downcomer apron' (i.e. for flow through the 'downcomer clearance'), h_t = total tray pressure drop for flow of the gas [= $h_d + h_l$, see Eq. (5.14)], $h_{db} = h_c + h_{ad} + h_t$ = downcomer backup; (b) 'backtrapping' or 'dumping' of liquid from a tray: Δ = hydraulic gradient, h_w = weir height, h_{ow} = crest height at weir outlet.

The hydraulic gradient may be large on a cross-flow tray if either or both of tower diameter and liquid load are large. The liquid flow path should not be kept too long in order to avoid maldistribution of liquid and vapour streams. A 'split flow' or 'double pass'configuration of the tray may be adopted in such a case. The liquid load is split into two or more equal parts. More than one downcomers are provided on a split-flow tray [Figure 5.8(a) and (b)]. The design of a two-pass tray is not very difficult since it is symmetric. But the challenge becomes critical with more than two passes of the liquid. Three-pass trays are rarely used since they are inherently asymmetric. Four-pass trays [Figure 5.8(c)] are more frequently used in relatively large diameter columns. A very important design requirement is 'balancing' of the liquid flows so that the gas

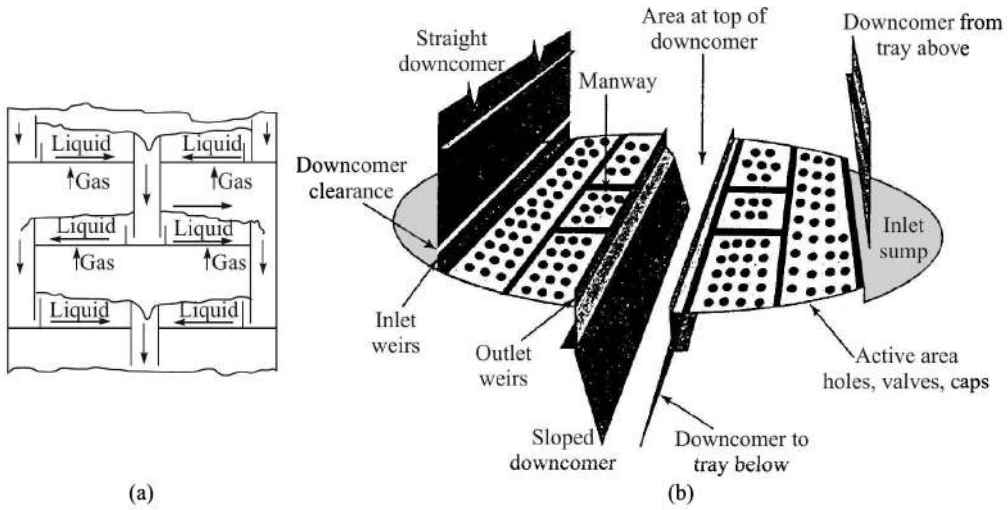


Figure 5.8(a) and (b) (a) Sketch of a 'split-flow' tray, two-pass; and (b) a two-pass sieve tray (Sloley, 1999) with sloped downcomers.

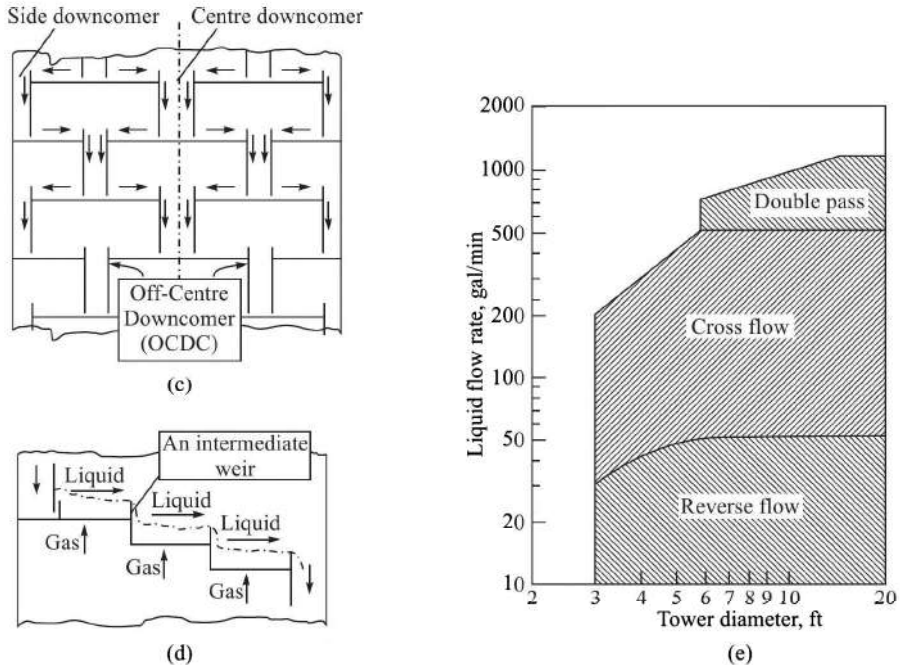


Figure 5.8(c), (d) and (e) (c) Schematic of a four-pass tray, (d) a 'cascade' or 'stepped' tray, and (e) the diagram for selection of liquid flow pattern.

and liquid flows remain uniform in all the four sections despite differences in the geometry. There are two approaches to this effect: (i) 'equal bubbling area' which means that each of the four sections should have the same number of openings for gas flow, and (ii) 'equal length of flow-path' which means that the liquid residence time is the same though the bubbling areas are not. Both these approaches have their own advantages and limitations. But in any case, the other hydraulic design parameters such as the weir shape [straight or swept-back, see Figure 5.13(c)] and height, and downcomer clearance should be properly adjusted to 'balance' the liquid flow and to ensure uniform performance of all the four sections. These issues have been discussed in a recent article (Pilling, 2005).

5.3.2 Weeping and Dumping

If a very small fraction of the liquid flows from a tray to the lower one through perforations or openings of the tray deck, the phenomenon is called 'weeping'. Weeping causes some reduction of the 'tray efficiency' because the liquid dripping down to the tray below through the perforations has not been in full contact with the gas or vapour. On the other hand, 'dumping' is an extreme case of leakage [shown in Figure 5.7(b)] through the tray deck if the vapour velocity is low and the vapour pressure drop across the tray is not sufficient to hold the liquid (Bennett and Kovac, 2000). In practice, a little bit of weeping may occur intermittently through sieve trays because of the instantaneous pressure imbalance.

5.3.3 Entrainment

When a gas bubbles through the liquid pool vigorously, droplets of liquid are formed in the vapour space by quite a few mechanisms including shearing action of the gas jet or rupture of the liquid film as a gas bubble bursts. Depending upon the size of a droplet, its velocity of projection and the drag force acting on it due to the gas velocity, the droplet may descend back into the liquid on the tray or may be carried into the tray above. The phenomenon of carry over of the suspended droplets into the upper tray is called 'entrainment'. The chances of entrainment are more if a droplet is small, if the gas velocity is large, or if the tray spacing is small.

Entrainment causes two major problems: (i) It leads to mixing of the entrained liquid from the lower tray with the liquid on the upper tray. This adversely affects the mass transfer driving force at the upper tray and reduces the 'tray efficiency'. (ii) The carry over of a substantial mass of liquid as droplets into the upper tray increases the liquid flow rate and downcomer load of that tray. In extreme cases, the second problem may lead to 'flooding' of the tower (as explained in Section 5.3.5). Entrainment is expressed as kg(droplets entrained)/s, kg per kg vapour, or kg per kg liquid flow, or kmol per kmol liquid flowing.

5.3.4 Flow Regimes on the Tray

The characteristics of gas–liquid dispersion on a tray depend upon the flow rates of the two phases, besides the tray type. A few 'flow regimes' may be identified (Lockett, 1986; Yanagi, 1990) in this connection [Figure 5.9(a)]. When the vapour (or the gas) rises through the liquid, there may be violent 'spray action' on one extreme or a 'quiescent flow' of the two-phase 'emulsion'

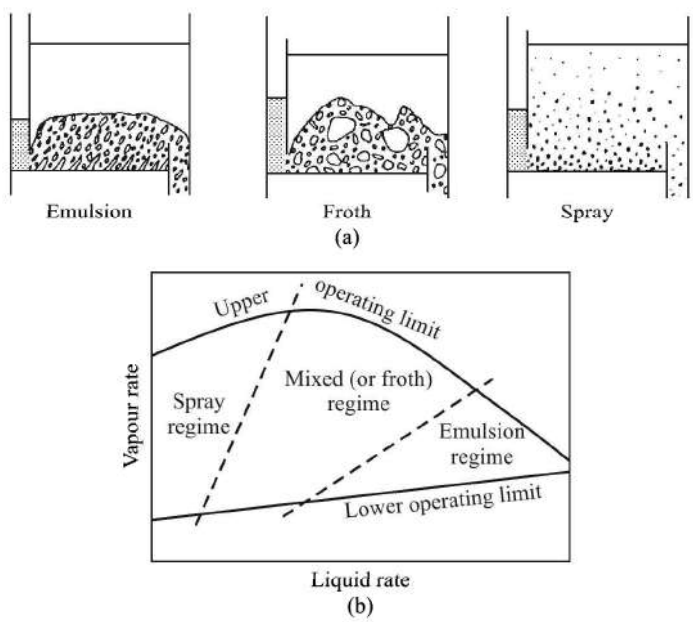


Figure 5.9 (a) Types of gas–liquid dispersion on a tray, and (b) flow regimes and operating limits of a sieve tray (qualitative).

(or dispersion) on the other extreme, depending upon the flow rates of the two phases. In between, there occurs a 'froth regime' [Figure 5.9(a)]. In addition to the flow regimes, the flow rates of the two phases (particularly of the vapour phase) also govern phenomena such as weeping, dumping, entrainment, and flooding. A *qualitative* description of the flow regimes and their effects are shown in Figure 5.9(b).

Spray regime

It occurs at low liquid rates and high vapour velocities. The vapour enters at a high velocity into a shallow liquid layer disintegrating the liquid into droplets which are projected up in the space between two consecutive trays. The gas becomes the continuous phase, and the liquid, the dispersed. This is called the 'spray regime'. If the vapour rate is very high and the tray spacing is not sufficient, the droplets cannot settle. These are rather carried by the gas into the upper tray causing an accumulation of liquid there and eventual 'flooding' (flooding is described in Section 5.3.5). This is more likely to occur in vacuum operation since the gas velocity remains very high. A change of flow regime from froth to spray is called *phase inversion* in a tray column since the gas changes from dispersed to continuous state.

Emulsion regime

This occurs if the gas flows through the liquid at a rather low rate. The vapour bubbles formed at the nozzles are 'sheared off' by the fast-moving liquid. The bubble size and its rise velocity remain low as a result. The liquid with the slowly rising entrapped bubbles appears like an 'emulsion' and hence the name of the regime.

Froth or mixed regime

In between the spray and emulsion flow regimes there exists a 'froth' or 'mixed flow' regime extending over a range of gas velocities. The liquid rate is large enough to prevent spraying but not too large to cause substantial shearing of the bubbles at the stage of formation. The froth regime is the preferred flow regime in industrial towers.

5.3.5 Flooding

Under normal operating conditions, an average liquid depth is maintained on a tray. 'Flooding' is an abnormal condition of excessive accumulation of liquid and simultaneous excessive pressure drop across the flooded tray. If accumulation of liquid on a tray continues, a part or eventually the entire column may be filled with the liquid. Flooding of a tray or a column may occur because of one or more of the following reasons (Kister, 1989; Yanagi, 1990).

Spray entrainment flooding

At a low liquid rate, the spray height increases as the vapour velocity is increased. The spray may eventually reach the downside of the tray above, causing a substantial carry over of the liquid droplets and leading to flooding of the tray and the column.

Froth entrainment flooding

Froth appears on a tray over a range of gas and liquid flow rates. Depending upon the flow rates of the phases and the tray spacing, the froth may almost fill the entire space between two

successive trays. Most of the liquid droplets suspended in the gas are then carried into the upper tray causing liquid accumulation thereon and eventual flooding of the tray. If the liquid is prone to foaming on aeration, flooding may occur even at lower gas velocities.

Downcomer flooding

When the liquid flow rate is high, the rate of flow of the gas–liquid dispersion into the downcomer also becomes high. If the downcomer cannot accommodate the dispersion load and the gas–liquid disengagement does not occur properly, the aerated frothy liquid fills the downcomer and finally backs up onto the tray above (Yanagi, 1990) leading to 'downcomer flooding'.

Let us refer to Figure 5.7(a). The static pressure exerted by the froth in the downcomer, expressed as 'clear liquid height', balances the sum of three terms (Yanagi, 1990): (i) the clear liquid height on the the tray (h_c); (ii) the head loss for liquid flow under the downcomer plate or the 'downcomer apron' (h_{ad}), which depends upon the downcomer clearance, the liquid flow rate and liquid properties; and (iii) the total gas pressure drop for flow through the liquid on the tray (h_t), which is the sum of dry tray pressure drop and the pressure drop for passage of the gas through the liquid. The sum of these three terms, all expressed as 'liquid head', is called the 'downcomer backup' (h_{db}) under stable tray operation, i.e.

$$h_{db} = h_c + h_t + h_{ad} \tag{5.1}$$

The froth height h_f over the tray floor in the downcomer is given by

$$h_f = \frac{h_{db}}{\phi_d} \tag{5.2}$$

where ϕ_d is the average relative froth density in the downcomer ($\phi_d = \rho_f/\rho_L$; ρ_f = froth density, and ρ_L = liquid density). Downcomer flooding occurs if the froth height h_f crosses the weir level of the upper tray, i.e.

$$h_f \geq S + h_w \qquad (S = \text{tray spacing; } h_w = \text{weir height}) \tag{5.3}$$

One strategy of overcoming the problem of downcomer flooding is to provide a larger tray spacing. Downcomer backup is further discussed in Section 5.5.2 and illustrated in Example 5.1.

5.3.6 The Performance Diagram

The performance of a particular type of tray depends upon the *relative* liquid and vapour throughputs. The effects of the phase flow rates on the performance of a sieve tray are qualitatively shown in Figure 5.10 (Yanagi, 1990). This is also called the *tray stability diagram*.

Let us assume that the tray operates at gas and liquid rates corresponding to point N located in the 'region of normal operation' in Figure 5.10. If we keep on increasing the gas throughput at a constant liquid rate, more liquid is entrained in the gas. At point E, the gas rate is high enough to cause excessive entrainment. The tray efficiency falls to an unacceptable level. At point F, the tray gets flooded and is inoperable.

Also, the tray does not operate satisfactorily if the gas rate is kept on decreasing. At a gas rate corresponding to point W, the gas pressure drop across the tray decreases to the extent that some liquid starts leaking through the tray holes. This is weeping of the tray as discussed before. As the gas rate is further reduced to point L, the weeping rate increases significantly causing

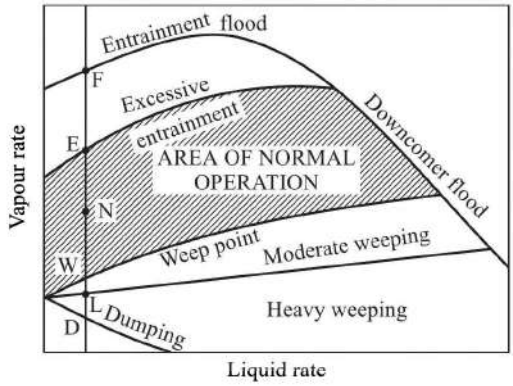


Figure 5.10 Performance diagram of a sieve tray.

deterioration of performance of the tray. The tray should be operated above this point. If the gas throughput is yet reduced to the point D, dumping of liquid starts. Under such a condition, most of the liquid passes through the tray holes and little, if any, flows into the downcomer.

The operable limits of a tray in respect of gas and liquid flow rates depend on the tray design, temperature and pressure conditions as well as the properties of the liquid and vapour phases.

5.3.7 Turndown Ratio

A column designed for a particular capacity may have to be operated at an enhanced or reduced capacity depending upon changes in the production rate in the plant due to various factors. It is, therefore, desirable that the trays have some degree of *flexibility* to operate over a range of throughput around the design capacity. Such flexibility is expressed in terms of the 'turndown ratio', which is defined as the ratio of the 'design vapour throughput' to the 'minimum operable throughput'.

Sieve trays have a low turndown ratio of about 2 (i.e. a sieve tray can normally be operated up to 50% of the design vapour throughput; at a still smaller vapour rate, severe weeping occurs). This value can be increased by reducing the fractional hole area. But this strategy is not very effective because of increased vapour pressure drop and reduced capacity. Valve trays normally have a turndown ratio of 4; bubble-cap trays have a still larger turndown ratio.

5.3.8 Tray Spacing

Tray spacing is the distance or gap between two consecutive trays in a column. An adequate tray spacing is important for quite a few reasons. The rate of entrainment (see Section 5.3.4) strongly depends upon the tray spacing. If a larger spacing is provided, most of the liquid droplets descend back to the tray reducing the entrainment. The column can operate at a greater superficial gas velocity and a smaller column diameter can be used for a given throughput. But the column height increases if the same number of trays have to be accommodated. Hence, a trade-off between a smaller column diameter and a larger height has to be struck.

Tray spacing varies over a pretty wide range of 8 to 36 inches. For a column 4 ft or larger in diameter, a tray spacing of 18 to 24 inches is adequate. If frequent inspection or maintenance is required, a larger tray spacing is convenient and advisable. A tray spacing of 30 inches or more is used in a vacuum column. In a big diameter column, the tray support beams restrict the movement of a workman on a tray for the purpose of cleaning and maintenance. So a larger spacing is necessary. If the liquid has a foaming tendency, a larger tray spacing should be allowed. A spacing less than 18 inches is not recommended for the common three tray types unless for any compelling reason.

5.3.9 Comparison of Bubble Cap and Sieve Trays

A qualitative comparison of the three commonly used trays in respect of capacity, efficiency, flexibility (turndown ratio), cost and a few other criteria is given in Table 5.2. It is to be noted that each of the criteria depends upon the particular system and operating conditions.

Table 5.2 Comparison of three common tray types (Kister, 1992)*

Parameter	*Bubble-cap tray*	*Sieve tray*	*Valve tray*
Capacity	Moderate	High	High to very high
Efficiency	Moderate	High	High
Entrainment	High (about three times that of a sieve tray)	Moderate	Moderate
Pressure drop	High	Moderate	Moderate
Turndown	Excellent (can operate at a very low capacity)	About 2 (not suitable to operate at variable loads)	4-5
Fouling tendency	High, tends to collect solids	Low	Low to moderate
Cost	High (about 2–3 times that of a sieve tray)	Low	About 20% more than sieve trays
Applications	Rarely used in new columns, may be used if low flow rates are anticipated	Most applications if turndown is not important	Preferred if a high turndown is anticipated
Share of market	About 5%	25%	About 70%

*Comparative assessment of trays should be done mainly on the basis of flexibility (turndown ratio), efficiency (discussed in Chapter 6), pressure drop and cost. Bubble-cap trays are more expensive and are not preferred for new installations. But they can satisfactorily operate at 50 to 120% of the design vapour velocity and 15 to 130% of the design liquid load.

5.4 A FEW MECHANICAL DETAILS

Mechanical construction of trays

It has been stated before that the trays are supported on tray-support rings welded to the shell. However, large trays are invariably made up of a few sections. Sometimes it may be necessary to take out a section through a manway for maintenance or replacement. The tray sections are screwed or bolted together to form the complete tray. The peripheral edge of a tray is bolted or clamped to the tray-support ring. Slotted bolt-holes in the ring are sometimes used for ease of assembling. If the tray and the shell have unequal expansions while operating at an elevated